

KHUSHI

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Engineering student with hands-on experience building AI-powered and agentic systems, including retrieval-augmented generation (RAG) pipelines, LLM tool-calling architectures, generative AI integrations, machine learning pipelines, and full-stack backend and Android development. Proficient in Python, FastAPI, Flask, Java, JavaScript, and SQL, with experience in LLM tooling (Ollama, Qwen, Groq), vector databases, Android Studio, Git/GitHub, and cloud fundamentals.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, SQL

AI/ML & LLM Tools: Retrieval-Augmented Generation (RAG), Hybrid Search (Semantic Embeddings + BM25), Vector Databases (ChromaDB), Ollama, Qwen, Groq API, Agentic AI / Tool-Calling Architectures, Prompt Engineering, Generative AI APIs, Machine Learning (Recommendation Systems), Pandas, NumPy

Mobile Development: Java, Android Studio, Android SDK

Backend & Frontend: FastAPI, Flask, Node.js, React.js, HTML5, CSS3, JavaScript, REST API Design

Databases: MySQL, PostgreSQL, SQLite, Firebase, Supabase, SQL Query Design

Tools & Platforms: Git, GitHub, Docker, AWS/Cloud Fundamentals, Linux, Jupyter, VS Code

EXPERIENCE

Python AI/ML Developer Intern, Solitaire Infosys, Mohali

Jun 2024 - Jul 2024

- Developed and tested Python-based backend modules within a structured software engineering environment, following modular design and version control practices with Git/GitHub.
- Built data-processing pipelines using Pandas and NumPy to clean, transform, and serve structured datasets; produced EDA reports and visualizations using Matplotlib and Seaborn, collaborating with senior engineers on documentation and delivery.

PROJECTS

AdmitAI - Agentic AI-Integrated College Admission Website (In Progress)

2026

- Designing and integrating a custom agentic AI chatbot into a live college admission website using Ollama (Qwen), Python, and FastAPI, enabling the LLM to autonomously select and chain 15+ custom tools for real-time course, fee, and eligibility queries.
- Building a lead-capture chat widget (HTML5, CSS3, JavaScript) with a counselor-facing admin dashboard, integrating Supabase (PostgreSQL); engineering reliability safeguards including fuzzy string matching and deterministic Python calculations to replace unreliable model-generated arithmetic.

Marginalia - AI-Powered Document Research Assistant (RAG-Based)

2026

- Built a full-stack Retrieval-Augmented Generation (RAG) application enabling users to upload documents and query them in natural language, combining a FastAPI backend, ChromaDB vector store, and Groq-hosted LLM inference for real-time, source-grounded answers, with a hybrid retrieval pipeline blending sentence-transformer embeddings and BM25 keyword search for improved context relevance.
- Designed a streaming chat interface using Server-Sent Events for real-time token-by-token responses, with page-aware document chunking, persistent multi-session chat history (SQLite), and inline citation tracking linking every answer back to its exact source and page.

Women Safety Application - Java-Based Android App (Android Studio)

Apr 2026 - Jun 2026

- Built a native Android application in Java using Android Studio, delivering real-time location tracking and emergency-alert features through a Python/Flask REST API backend with JSON API contracts.
- Integrated geospatial data processing with threshold-based trigger logic, applying modular MVC-style design across app and backend for maintainability and scalability.

Smartflix - ML-Based Movie Recommendation System

Jun 2025 - Jul 2025

- Designed a full-stack machine learning-based recommendation application with a Python backend REST API and an HTML/CSS/JavaScript frontend.
- Engineered ML pipelines processing a 10,000+ entry metadata corpus using cosine-similarity and matrix-factorization algorithms to generate personalized movie recommendations.

EDUCATION

B.Tech, Computer Engineering, Amritsar Group of Colleges, Amritsar, Punjab

Aug 2023 - May 2027

Relevant Coursework: Software Engineering, Data Structures & Algorithms, Web Technologies, Database Systems, Cloud Computing (IIT Kharagpur), Operating Systems

CERTIFICATIONS

Cloud Computing - IIT Kharagpur (NPTEL), 2026 | Software Engineering - IIT Kharagpur (NPTEL), 2025 | Intro to Machine Learning - IIT Madras (NPTEL), 2026 | Deep Learning for Developers - Infosys Springboard, 2026

HACKATHONS & ACHIEVEMENTS

- 2nd Place, MindSpace Hackathon - Designed and shipped a working web-based AI solution within a constraint, demonstrating rapid full-stack prototyping and cross-functional collaboration.
- Participant, UIDAI Aadhaar Hackathon - Contributed to a government-sponsored hackathon on data privacy and identity systems, applying AI/ML concepts.